Ahmedabad University | Theoretical AI

Bayesian Personalized Recommendation using Latent Trait Models & Expectation Propagation

Results and Analysis

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https://github.com/Jyotier2006/M2_g4_th_apl_recommend-sys_team

Abstract

We compare assumed-density filtering (ADF), repeated expectation propagation (EP), matrix factorization and a Gaussian Markov random field (GMRF) prior for movie-rating prediction. Across five MovieLens 1M splits, RMSE is 0.8977 ± 0.0035 for ADF, 0.8643 ± 0.0037 for EP5 and 0.8548 ± 0.0029 for EP10. GMRF-MAP reaches 0.8499 ± 0.0023, versus 0.8506 ± 0.0021 for its matching graph-free MAP baseline. The isolated graph gain is small; four split-specific user-bootstrap intervals exclude zero and one includes it. Larger gains occur under missing or corrupted training ratings. Calibration, cold-start and sensitivity tests reveal limits to these benefits rather than universal superiority.

1 Problem and dataset

The task is to predict an unobserved 1–5 movie rating and its probability distribution. Shared user/movie latent variables transfer evidence between interactions. We examine whether revisiting ratings improves approximate Bayesian inference, and whether explicit movie dependencies improve a matching factorization model.

MovieLens 1M contains 1,000,209 ratings from 6,040 users on 3,706 rated movies; the metadata catalogue contains 3,883 movies [1, 2]. The official archive and input files are SHA-256 verified. Density is 4.468%; an absent interaction is unobserved, not a zero rating.

Field	Meaning	Code representation/use
UserID, MovieID	Entity identities	Zero-based arrays u, i; raw mappings retained
Rating	Observed integer 1–5	Target array r
Timestamp	Unix observation time	Audited; no temporal predictor
Title, Genres	Movie name and labels	Title for identification; 18-column multi-hot genres for graph
Gender, Age, Occupation, Zip-code	Demographic categories	Parsed/audited; not predictors

Table 1: Dataset variables and their actual uses. IDs are categorical keys.

2 Latent-trait probabilistic model

For user u and movie i, vectors $s_u, t_i \in \mathbb{R}^K$ and biases b_u, b_i generate affinity and an ordinal rating:

$$a_{ui} = \mu + b_u + b_i + s_u^T t_i, \quad y_{ui} \mid a_{ui} \sim \mathcal{N}(a_{ui}, \beta^2), \tag{1}$$

$$R_{ui} = r \iff c_{r-1} < y_{ui} \le c_r, \quad c = (-\infty, 1.5, 2.5, 3.5, 4.5, \infty). \tag{2}$$

The offset μ is the training mean. Default independent priors are $s_{uk} \sim \mathcal{N}(m^0_{uk}, 0.25)$, $t_{ik} \sim \mathcal{N}(n^0_{ik}, 0.25)$ and $b_u, b_i \sim \mathcal{N}(0, 1)$. Seeded prior means sampled from $\mathcal{N}(0, 0.1^2)$ break bilinear symmetry. For $\Theta = (S, T, b_U, b_I)$,

$$p(\Theta, R_{\mathcal{D}}) = p(\Theta) \prod_{(u,i)\in\mathcal{D}} \left[\Phi\left(\frac{c_{R_{ui}} - a_{ui}}{\beta} \right) - \Phi\left(\frac{c_{R_{ui}-1} - a_{ui}}{\beta} \right) \right]. \tag{3}$$

Ratings are conditionally independent given their relevant parameters. Shared users/movies induce dependence after marginalization. Preferences are static; exposure mechanisms and causal effects are not modelled.

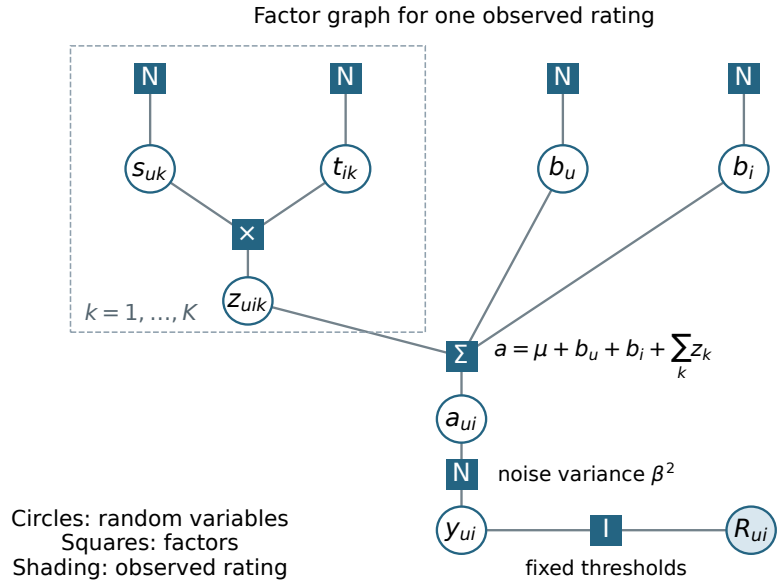


Figure 1: One rating's factor graph. Circles are variables and squares are factors. Trait products represent compatibility; the sum adds biases; noise and thresholds turn affinity into the observed star.

At $K = 8$, each user/movie has eight unnamed numeric traits and a bias: $(6040 + 3706)(8 + 1) = 87,714$ persistent scalar latent parameters. A coordinate is not automatically a genre. This excludes observations, auxiliary variables, hyperparameters and EP sites.

3 Learning and probabilistic prediction

3.1 ADF and repeated EP

Exact posterior inference is intractable for the coupled bilinear ordinal model. Hybrid message passing uses VMP at products and EP ordinal moment matching [4]. Scalar beliefs are Gaussian, stored through precision $\lambda = 1/v$ and weighted mean $\eta = m/v$. Before revisiting rating site n, EP forms a cavity:

$$\lambda^{\setminus n} = \lambda - \lambda_n, \qquad \eta^{\setminus n} = \eta - \eta_n. \tag{4}$$

A local update then replaces the old site. ADF processes each rating once and discards its site; with identical priors and order, ADF equals EP1 exactly. EP2–EP10 mean passes through the same ratings. Repeating ADF without cavity removal would double-count evidence.

For incoming score $\mathcal{N}(m, v)$, set $\alpha = (c_{r-1} - m)/\sqrt{v}$, $\delta = (c_r - m)/\sqrt{v}$ and $Z = \Phi(\delta) - \Phi(\alpha)$. Truncated moments are

$$\widehat{m} = m + \sqrt{v}\,\frac{\phi(\alpha) - \phi(\delta)}{Z}, \tag{5}$$

$$\widehat{v} = v\left[1 + \frac{\alpha\phi(\alpha) - \delta\phi(\delta)}{Z} - \left(\frac{\phi(\alpha) - \phi(\delta)}{Z}\right)^2\right]. \tag{6}$$

Gaussian division provides the outgoing message. Reverse VMP through a product uses the other trait's second moment $\mathbb{E}[t_k^2] = v_{t_k} + m_{t_k}^2$. Local damping is 0.7, with at most ten iterations and tolerance 10^{-4}. Outer replacement uses weight 1 on pass one and 0.5 thereafter by default.

3.2 Five-star prediction

Under factorized Gaussian beliefs,

$$M_k = m_{s_k} m_{t_k}, \quad V_k = v_{s_k} v_{t_k} + v_{s_k} m_{t_k}^2 + v_{t_k} m_{s_k}^2, \tag{7}$$

$$m_y = \mu + m_{b_u} + m_{b_i} + \sum_k M_k, \quad v_y = \beta^2 + v_{b_u} + v_{b_i} + \sum_k V_k, \tag{8}$$

$$p_r = \Phi\left(\frac{c_r - m_y}{\sqrt{v_y}}\right) - \Phi\left(\frac{c_{r-1} - m_y}{\sqrt{v_y}}\right), \quad \widehat{R} = \sum_{r=1}^{5} r p_r. \tag{9}$$

Product moments are exact under factorized beliefs; making the total score Gaussian is an additional approximation. Probabilities are floored at 10^{-15} and renormalized. Expected stars and the most probable star are different decisions, used for RMSE and exact-star accuracy respectively.

Nine numerical/invariant checks pass, including kernel equivalence, ADF–EP1 equality, site accounting, truncation moments, valid graph precision and nonincreasing MAP energy. Bilinear quadrature exposes variance errors of about 10% and lost covariance. A separate 12-variable Gaussian diagnostic compares exact inference, Gauss–Seidel and Gibbs sampling; these are small-model validation results, not MovieLens recommendation scores.

4 Gaussian Markov movie network

4.1 Constructing the network

Undirected edges encourage related movie vectors to remain similar. Five alternating regularized bias updates produce training residuals $e_{ui} = R_{ui} - \mu - b_u - b_i$. Normalized residual columns yield positive cosine similarities, shrunk by $n_{ij}/(n_{ij} + 20)$ and discarded when fewer than five training users rated both movies. Default affinities mix 90% residual similarity with 10% cosine similarity of multi-hot genres.

Each movie retains its top 20 positive neighbours. The union is symmetrized by maximum weight. For symmetric affinity A, set $W = D_A^{-1/2} A D_A^{-1/2}$, recompute $D_{ii} = \sum_j W_{ij}$ and $L = D - W$. Validation/test ratings never construct edges. The graph averages 56,508 undirected edges, degree 30.50 and density 0.823%; symmetrization can exceed 20 neighbours. Estimated edges are held fixed, not given a structure posterior.

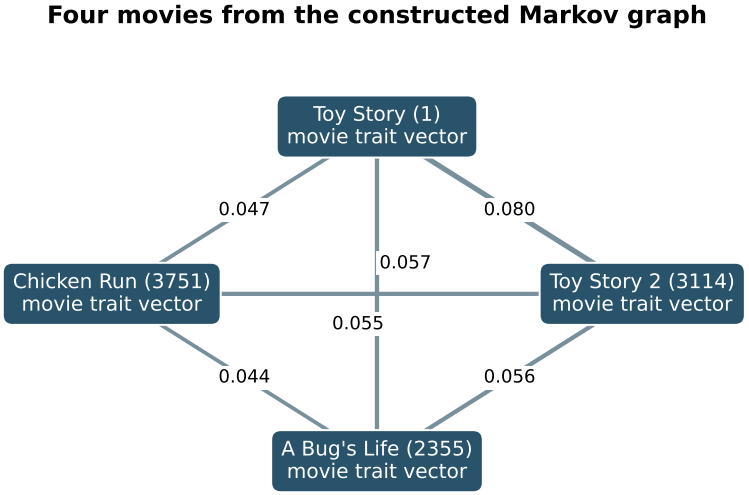


Figure 2: Actual seed-42 subgraph. Nodes are latent movie vectors; labels are normalized edge weights. Similarity is not a causal relationship.

4.2 Prior and MAP objective

For vectors Q_i, the proper Gaussian Markov prior is

$$p(Q \mid W) \propto \prod_i e^{-\lambda \|Q_i\|^2/2} \prod_{\{i,j\} \in E} e^{-\gamma w_{ij} \|Q_i - Q_j\|^2/2}, \quad J = \lambda I + \gamma L. \tag{10}$$

For $\lambda > 0$, $x^T J x = \lambda \|x\|^2 + \gamma \sum_{\{i,j\}} w_{ij}(x_i - x_j)^2 > 0$ for $x \neq 0$. Precision zeros encode conditional independence in the prior, not marginal independence. The conditional mean of Q_{ik} is $\gamma \sum_j w_{ij} Q_{jk}/(\lambda + \gamma D_{ii})$, and variance is $1/(\lambda + \gamma D_{ii})$.

With $f_{ui} = \mu + b_u + b_i + P_u^T Q_i$, Gaussian numeric-rating likelihood gives

$$\mathcal{E} = \tfrac{1}{2} \sum_{\mathcal{D}} (R_{ui} - f_{ui})^2 + \tfrac{\lambda}{2}(\|P\|_F^2 + \|Q\|_F^2) + \tfrac{\lambda_b}{2}(\|b_U\|^2 + \|b_I\|^2) + \tfrac{\gamma}{2} \operatorname{tr}(Q^T L Q). \tag{11}$$

Alternating user/movie block solves add $\gamma D_{ii} I$ to a movie trait block and $\gamma \sum_j w_{ij} Q_j$ to its right-hand side. Each exact block update cannot increase energy; joint optimization remains nonconvex. Defaults are $K = 8$, $\lambda = \lambda_b = 10$ and 25 epochs.

5 Experimental protocol and KPIs

Seeds 42–46 each define 800,167 training, 100,020 validation and 100,022 test ratings through one random permutation. Every method uses the same split within a seed. Seeds control splitting/order and initialization. Fitted statistics use training ratings; validation selects settings; test metrics never select a model. These are random-interaction rather than temporal splits.

Global/user/item predictors use smoothed empirical star frequencies. MF-SGD uses eight traits, learning rate 0.007, decay 0.98, regularization 0.03 and 25 epochs. Its regularization convention differs from MAP. ADF and EP1–5/10 use common priors and the ordinal model. MAP tests $\gamma \in \{0, 1, 10, 100\}$; validation clipped-score RMSE selects checkpoints and strength. Selection can choose zero, disabling the graph.

MAP probabilities use a validation-NLL-fitted Gaussian residual variance integrated between the same rating thresholds. This models residual spread, not posterior parameter uncertainty. Comparing GMRF with EP changes likelihood and fitting method too. **The isolated graph comparison is GMRF-MAP versus the identical MAP model with $\gamma = 0$.**

5.1 Metrics and statistical variation

For observed R_n and expected $\widehat{R}_n = \sum_r r p_{nr}$,

$$\text{RMSE} = \sqrt{N^{-1} \sum_n (\widehat{R}_n - R_n)^2}, \qquad \text{MAE} = N^{-1} \sum_n |\widehat{R}_n - R_n|, \tag{12}$$

$$\text{NLL} = -N^{-1} \sum_n \log p_{n,R_n}, \qquad \text{Brier} = N^{-1} \sum_n \sum_r (p_{nr} - \mathbf{1}\{r = R_n\})^2. \tag{13}$$

ECE uses ten equal-width top-confidence bins; exact-star accuracy compares $\arg\max_r p_r$ with the observed rating. Lower RMSE, MAE, NLL, Brier and ECE are better. RMSE is in star units, not a percentage accuracy.

Tables give mean ± sample SD. Within each fitted split, 2,000 paired bootstrap replicates resample users and retain all their test ratings for both models. Intervals condition on fitted models, omit retraining uncertainty and shared-item dependence, and do not make overlapping splits independent. No naive independent-seed significance claim is made.

5.2 Controlled stress tests

Three seeds (42–44) test nested training retention of 25%/50%, replacement of 10%/20% of training labels by a uniformly selected different star, top-5/top-50 graph density, rating-only/genre-only graphs, and cold-start. Validation/test labels remain clean and graphs are rebuilt from modified training data. Strength 10 is retained from clean-validation comparisons and tested against zero; validation still selects epochs.

Cold-start removes all training rows for 10% of movie or user identities; test scores use only those entities' original test rows. Genres remain available, and checkpoint selection uses the full original validation split. EP5 also varies observation variance $\beta^2 \in \{0.5, 1, 2\}$, prior variance, damping and order. These noise assumptions differ from corrupting dataset labels.

6 Main predictive results

Method	RMSE	MAE	NLL
Global	1.1164 ± 0.0021	0.9334 ± 0.0018	1.4552 ± 0.0015
User mean	1.0342 ± 0.0021	0.8303 ± 0.0020	1.3461 ± 0.0016
Item mean	0.9786 ± 0.0026	0.7826 ± 0.0021	1.3209 ± 0.0028
MF-SGD	0.8683 ± 0.0024	0.6906 ± 0.0023	1.2214 ± 0.0025
ADF / EP1	0.8977 ± 0.0035	0.7098 ± 0.0030	1.2519 ± 0.0033
EP3	0.8740 ± 0.0036	0.6894 ± 0.0026	1.2283 ± 0.0033
EP5	0.8643 ± 0.0037	0.6808 ± 0.0025	1.2189 ± 0.0035
EP10	0.8548 ± 0.0029	0.6721 ± 0.0018	1.2098 ± 0.0024
MF-MAP (no graph)	0.8506 ± 0.0021	0.6753 ± 0.0017	1.2012 ± 0.0021
GMRF-MAP	0.8499 ± 0.0023	0.6752 ± 0.0019	1.2007 ± 0.0024

Table 2: Expected-rating test errors, mean ± sample SD over five seeds.

Method	Brier	ECE	Star accuracy
MF-SGD	0.6554 ± 0.0011	0.0397 ± 0.0010	46.04 ± 0.16%
ADF / EP1	0.6681 ± 0.0012	0.0451 ± 0.0010	44.13 ± 0.13%
EP3	0.6590 ± 0.0012	0.0537 ± 0.0017	45.20 ± 0.18%
EP5	0.6552 ± 0.0012	0.0566 ± 0.0009	45.59 ± 0.16%
EP10	0.6516 ± 0.0007	0.0608 ± 0.0012	46.04 ± 0.21%
MF-MAP (no graph)	0.6467 ± 0.0007	0.0383 ± 0.0018	46.98 ± 0.15%
GMRF-MAP	0.6468 ± 0.0009	0.0387 ± 0.0022	46.96 ± 0.19%

Table 3: Probability quality and exact-star accuracy (percent), five seeds.

Repeated EP improves RMSE from 0.8977 to 0.8643 at five passes and 0.8548 at ten. Revisiting evidence helps, but more passes increase cost and do not prove convergence. GMRF reaches 0.8499 versus graph-free MAP's 0.8506: most of its advantage over EP5 is already present without the graph. MF-SGD remains an inexpensive competitive baseline.

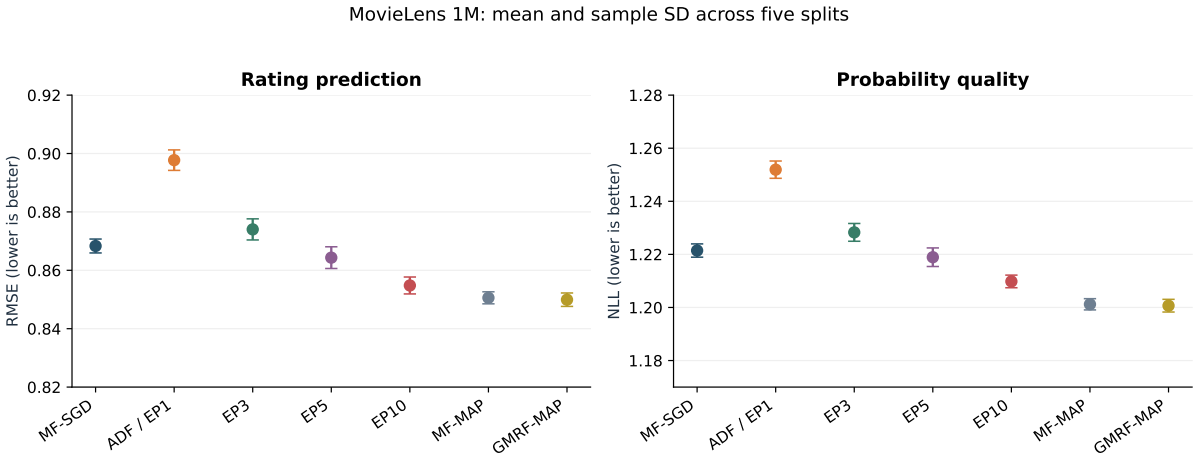


Figure 3: Prediction error and probability quality; error bars show sample SD.

Secondary clipped-score RMSE is 0.8650 for SGD, 0.8478 for MF-MAP and 0.8465 for GMRF-MAP. These use a different decoder and are not mixed with the primary expected-rating results.

7 Graph effect and ablation

All five clean-validation searches select $\gamma = 10$. The isolated graph gain is 0.000644 RMSE, about 0.076%. Every split's point estimate improves, but only four user-bootstrap intervals exclude zero; seed 43 includes it. This supports a modest, qualified benefit rather than universal superiority.

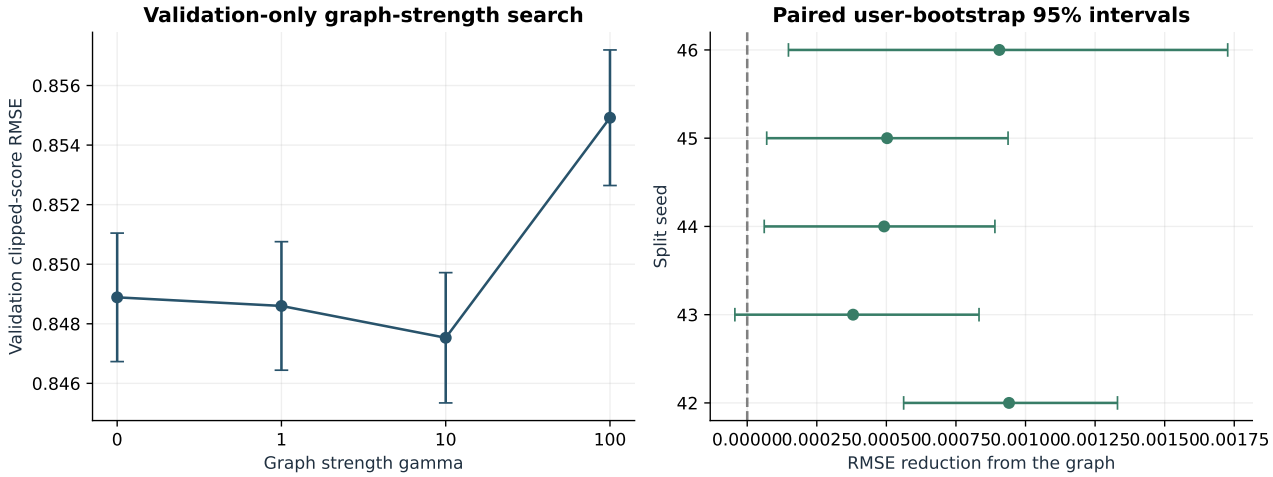


Figure 4: Validation-strength search and conditional paired 95% intervals for graph-free minus graph RMSE. Positive differences favour the graph.

Graph condition	MF-MAP RMSE	GMRF-MAP RMSE
main	0.8507 ± 0.0028	0.8501 ± 0.0031
ratings_only	0.8507 ± 0.0028	0.8499 ± 0.0031
genres_only	0.8507 ± 0.0028	0.8510 ± 0.0031
graph_top_5	0.8507 ± 0.0028	0.8497 ± 0.0031
graph_top_50	0.8507 ± 0.0028	0.8505 ± 0.0031

Table 4: Graph variants at fixed strength, mean ± SD over three seeds.

Ratings-only edges give RMSE 0.8499, mixed edges 0.8501 and genre-only edges 0.8510, versus 0.8507 without a graph on these three splits. Genre labels alone are too coarse for this warm-interaction setting; they need not imply similar audience preferences. Adding metadata is not automatically beneficial.

Top-5 neighbours give 0.8497 and top-50 give 0.8505. Strength 100 also worsens validation error. Extra or strong connections can oversmooth distinct traits. These exploratory variants do not replace the main top-20 configuration based on test scores. Graph ECE slightly worsens from 0.03835 to 0.03868; the RMSE gain is not uniformly improved probability quality.

8 Missing data, corrupted ratings and cold-start

Condition	ADF	EP5	MF-MAP	GMRF-MAP
main	0.8980 ± 0.0049	0.8633 ± 0.0040	0.8507 ± 0.0028	0.8501 ± 0.0031
retain_0.25	0.9217 ± 0.0041	0.9199 ± 0.0051	0.9178 ± 0.0042	0.9100 ± 0.0038
retain_0.5	0.9102 ± 0.0039	0.8920 ± 0.0049	0.8810 ± 0.0039	0.8775 ± 0.0041
noise_0.1	0.9129 ± 0.0036	0.8910 ± 0.0036	0.8827 ± 0.0029	0.8797 ± 0.0030
noise_0.2	0.9386 ± 0.0030	0.9326 ± 0.0022	0.9317 ± 0.0023	0.9248 ± 0.0018
cold_items	1.0703 ± 0.0079	1.0781 ± 0.0068	1.0622 ± 0.0060	1.0545 ± 0.0077
cold_users	0.9837 ± 0.0137	0.9861 ± 0.0146	0.9935 ± 0.0115	0.9916 ± 0.0117

Table 5: Expected-rating RMSE, three-seed mean ± SD. Cold-start rows use restricted test populations and are not directly comparable with the clean row.

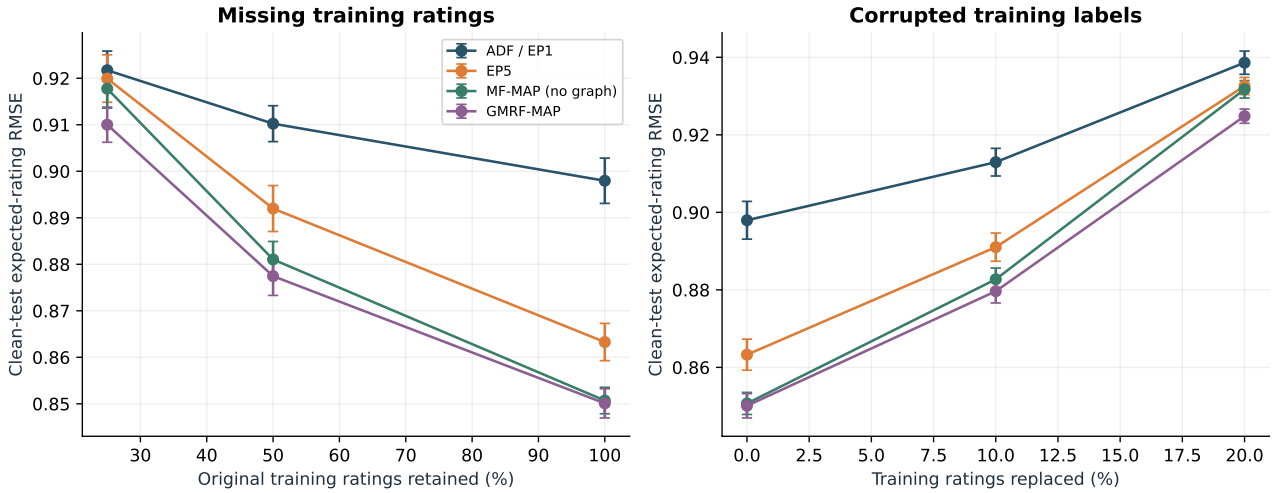


Figure 5: Effect of less available or less reliable training ratings.

At 25% retention the graph improves MF-MAP from 0.9178 to 0.9100; with 20% corrupted labels, from 0.9317 to 0.9248. These exceed the clean-data gain, consistent with reducing estimation variance by sharing evidence. Performance still degrades overall. Uniform missingness/corruption do not establish robustness to real nonrandom exposure or every user-error mechanism.

For unseen movies the graph improves MF-MAP from 1.0622 to 1.0545, but error remains high. Genre connections transfer neighbouring traits; the movie's own bias cannot be estimated. For unseen users, GMRF gives 0.9916 versus ADF's 0.9837: a movie graph cannot personalize a user with no observed preferences. EP5 also loses to ADF for unseen movies (1.0781 versus 1.0703). Extra inference passes cannot guarantee improvement when useful evidence is absent.

With only 25% of training ratings retained, EP5's advantage over ADF is small; seed 43 is slightly worse. Repeated approximate inference may reinforce weak or misspecified observations. Stress-test comparisons are descriptive results across three seeds, not independent-dataset significance tests.

9 Probability quality and sensitivity

EP5/EP10 improve NLL and Brier over ADF, but ECE worsens from 0.0451 to 0.0566 and 0.0608. Better rating error does not establish calibrated confidence. GMRF has better NLL here, but its probability wrapper omits parameter uncertainty.

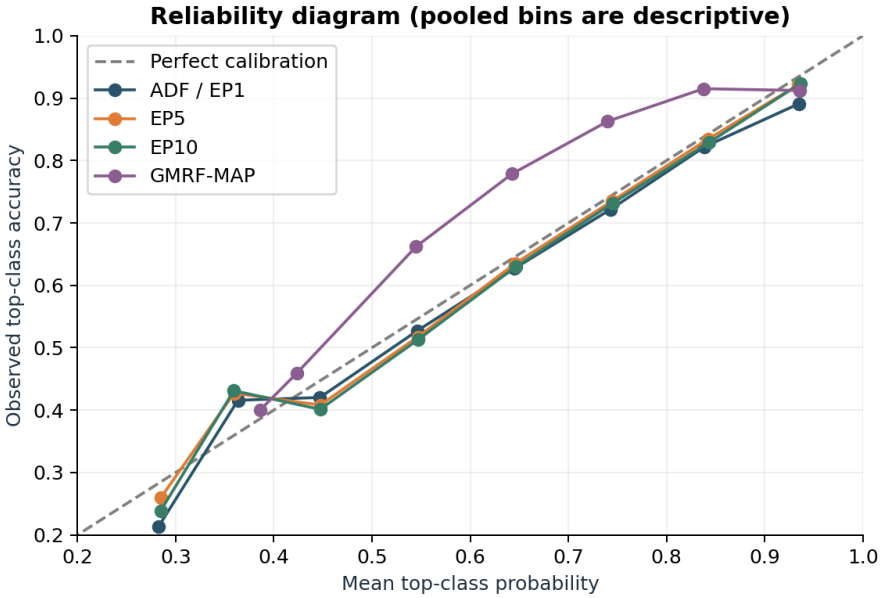


Figure 6: Top-class reliability. Overlapping split bins are pooled descriptively.

Noise variance	RMSE	NLL	ECE
0.5	0.8691 ± 0.0036	1.2404 ± 0.0054	0.0339 ± 0.0012
1.0	0.8643 ± 0.0037	1.2189 ± 0.0035	0.0566 ± 0.0009
2.0	0.8743 ± 0.0030	1.3154 ± 0.0020	0.0856 ± 0.0012

Table 6: EP5 observation-variance sensitivity, five seeds.

Validation NLL selects $\beta^2 = 1$ in every seed. Halving variance improves ECE to 0.0339 but worsens NLL to 1.2404 and RMSE to 0.8691. Doubling it worsens all three. Reducing assumed noise is therefore not a general accuracy improvement. Nominal 90% intervals cover 98.22% for EP5 and 97.41% for GMRF, with widths 3.82 and 3.60 star categories. Broad discrete intervals should not be called well calibrated merely because coverage is high.

Fixed prior, damping and order tests use three seeds. Outer damping 1 gives EP5 RMSE 0.8529 versus default EP5's 0.8633 and GMRF's 0.8501 on those same seeds. Schedule tuning closes much of the gap, further limiting attribution to graph structure. This exploratory variant does not replace the five-seed main method using test outcomes. Full sensitivity tables and validation trajectories accompany the code.

10 Error analysis and computational trade-offs

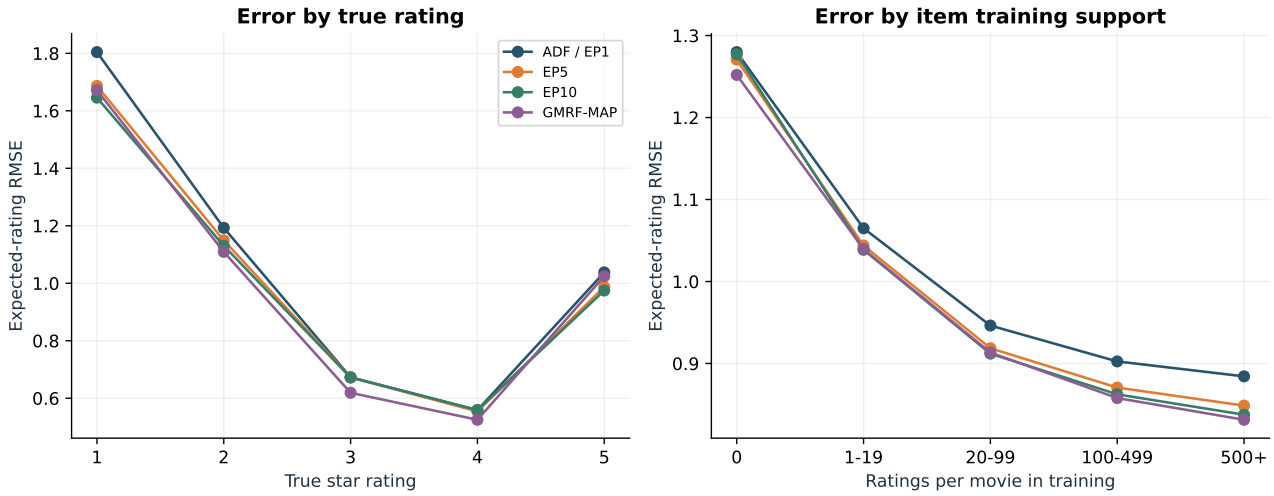


Figure 7: Errors by true star and movie support; pooled observations overlap.

GMRF's five-star RMSE is 1.0245 versus EP10's 0.9748 despite its better overall RMSE. Gains are stronger near three/four stars. Expected-rating predictions and regularization shrink extremes: low ratings are overpredicted and high ratings underpredicted. Low-support movies are harder for all methods. Only 83 pooled observations have naturally zero movie training support, so the controlled cold-start experiment provides stronger evidence.

Mean fit times are 3.33 s for ADF, 18.29 s for EP5, 35.99 s for EP10, 1.38 s for MF-SGD and 14.48 s for the selected GMRF candidate. Graph building adds 4.31 s and the full MAP search costs 66.37 s. Selected-candidate time alone cannot establish a faster graph workflow. JIT compilation is excluded. MAP/SGD fit timings include validation checks and scale fitting; EP fitting excludes validation scoring. Sequential CPU kernels use one numerical thread; versions, configurations and logs accompany the results.

For N ratings, K traits, L local iterations and T passes, EP training costs $O(TNL(K+1))$. Site storage is $32N(K+1)$ bytes: 219.77 MiB here. MAP block solves cost $O(T[N(K+1)^2+(U+I)(K+1)^3+|E|K])$, with $O(|E|)$ graph storage. More EP passes increase time rather than the number of sites. At pass 10, mean RMS parameter change remains 0.01564 despite no skipped main EP updates. Local numerical stability does not prove outer convergence.

11 Research positioning, limitations and conclusion

Matchbox, Microsoft Research, WWW 2009. Stern, Herbrich and Graepel [4] combine latent traits, metadata, feedback models and hybrid EP/VMP, with online ADF and temporal extensions. Their evaluation addresses recommendation, cold-start and scalability. This implementation is a static restricted variant with fixed thresholds and its own schedule; the full system and Infer.NET are not reproduced.

PreRec, AWS AI Labs affiliation, WSDM 2024. Lin et al. [3] use hierarchical Bayesian modelling, neural encoders, domain/popularity variables and causal debiasing for cross-domain recommendation. MAP training and cross-market/platform ranking evaluate zero-/few-shot transfer. Those representations, interventions and tasks are not implemented here; its scores are not comparable with this single-domain rating benchmark.

The evidence positions the project as an inference-and-regularization study, not a new algorithm or state-of-the-art claim. No true movie-dependency graph is available, so SHD/SID and structural recovery scores would be inappropriate.

11.1 Limitations and conclusion

Random-interaction splits do not measure future deployment. EP factorization loses correlations; MAP omits parameter uncertainty and treats graph edges as fixed. Searches are finite, epochs may reach the budget boundary, and optimizer regularization conventions differ. Artificial missingness/noise do not model all real user behaviour. Bootstrap intervals condition on fitted models and resample users only. User-specific thresholds, temporal dynamics and causal or cross-domain evaluation remain outside scope.

Repeated EP improves rating error at clear time and memory cost, without uniformly improving calibration. GMRF reaches 0.8499 ± 0.0023 RMSE, but its graph-free counterpart already reaches 0.8506 ± 0.0021. Its isolated gain is modest, with split-dependent statistical support. Larger gains under sparse or corrupted training support the explanation that related movies share information and reduce estimation variance. Genre-only, dense-graph and unseen-user failures show the limits. A network provides conditional benefit, not a guaranteed improvement for every recommender.

Reproducibility and accountability. Source, seeds, raw metrics, checks and logs accompany the report. Generative AI assisted implementation, execution, analysis and writing; team members remain responsible for checking results and confirming their actual contributions.

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